

Sharvin Neve

AI Systems & Inference Infrastructure Engineer

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Education

NMIMS Mukesh Patel School of Technology Management & Engineering, Mumbai, MBA Tech in Computer Engineering

2023 – 2028

- Integrated B.Tech + MBA (Computer Engineering) | CGPA: 3.4/4.0

- Coursework: **Operating Systems**, Computer Networks, Data Structures & Algorithms, OOP (C++), DBMS, Digital Electronics

Technical Skills

Languages: C, C++ (C++17/20, Object-Oriented Design), **Python**, **CUDA**, TypeScript, SQL, Bash/Shell Scripting

AI Systems & Inference: vLLM, **PagedAttention**, Continuous Dynamic Batching, AWQ / Marlin W4A16 GEMM, Prefix Caching (APC), PyTorch

Kernel & Embedded: RTEMS RTOS (Space-grade), POSIX, **Linux** Kernel / Device Drivers, **Raspberry Pi (ARM)**, Systemd, **GPIO**, IPC

Media & Cloud: **WebRTC** (LiveKit, SRS), **FFmpeg**, **OpenCV**, Cloudflare Tunnels, Docker, Git/GitHub, gRPC, **AWS** (Cloud Architecting)

Experience

Open Source Kernel Contributor, RTEMS Project – Remote

Feb 2026 – present

- Contributed upstream patches to the **RTEMS** space-grade real-time operating system kernel (deployed on ESA and NASA missions).
- Hardened watchdog timer validation logic and POSIX clock boundary checks across kernel core routines (coretodcheck.c, clock.h).
- Investigated 34-bit timer rollover constraints up to year 2400, eliminating edge-case integer overflows in real-time POSIX scheduling.

AI Systems Engineer Intern, Aspire Vision – Remote

Aug 2025 – Feb 2026

- Engineered real-time edge video streaming pipelines for an industrial smart-helmet safety system, achieving sub-300ms glass-to-glass latency over **WebRTC** / HTTP-FLV using **FFmpeg** and local SRS Docker containers on **Raspberry Pi (ARM)**.
- Configured zero-trust outbound Cloudflare Tunnels (cloudflared), bypassing carrier-grade NAT across 4 industrial field sites without opening inbound firewall ports.
- Designed offline resilience daemons managing physical **GPIO** camera triggers, camera hardware arbitration across **Linux** processes, and automated cloud sync (18GB footage processed, 14,750+ telemetry events).

Embedded Systems Intern, Entice Engineering – India

May 2025 – Aug 2025

- Built a **Python**-based video capture and GPS metadata tagging pipeline on embedded **Raspberry Pi** hardware for industrial field data collection.
- Interfaced **GPIO** sensors and optimized **OpenCV** frame processing routines under tight embedded memory and thermal limits.

Projects

High-Throughput LLM Serving & Batching Engine

- Designed an empirical continuous dynamic batching benchmark harness for Qwen 2.5 7B on an Ampere 12GB GPU (RTX 3060), scaling throughput from 57.2 tok/s to 592.0 tok/s (10.35x throughput scaling) across 20 concurrent streams.
- Implemented AutoAWQ Marlin W4A16 GEMM kernels to reserve 3.6 GiB VRAM for a 67,440-token **PagedAttention** KV-cache pool; achieved an 87.1% Radix-Tree Prefix Cache hit rate, reducing P50 TTFT to 38.7 ms.

Streaming-Rpi — Production Edge Video Streaming Stack

- Built and open-sourced a wearable camera streaming stack on **Raspberry Pi** with hardware-accelerated H.264 ingestion, SRS Docker container, and Next.js 16 / React 19 control plane dashboard with sub-300ms latency.

Certifications and Achievements

- AWS Academy Graduate** — Cloud Architecting (**AWS** Training Badge, 2026)
- AWS Cloud Quest: Cloud Practitioner** — **AWS** Training Badge
- Smart India Hackathon — National Finalist
- NPTEL Elite Certificate — Design Practices for Intelligent Product Design (IIT Kanpur)